

Predicting Student Dropout Through Staged Enrollment-Survival Modelling: A Behavioural Analytics Approach

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Abstract

This report presents a data-driven investigation into the factors governing student attrition in higher education. Departing from static, single-snapshot prediction, we adopt a *staged enrollment-survival* framework that decomposes the academic year into four natural checkpoints aligned with tuition-fee deadlines and assessment milestones. At each stage, the surviving cohort is modeled with an algorithm whose complexity is commensurate with the richness of the available feature set — from Logistic Regression at the earliest, data-sparse checkpoint to a Stacking ensemble at the final, feature-dense stage. All four stage-specific classifiers achieve ROC-AUC scores exceeding 0.97 under stratified 5-fold cross-validation, demonstrating that the staged approach not only respects the structural missingness inherent in longitudinal enrollment data but also yields highly actionable early-warning signals. A complementary SHAP-based interpretability analysis reveals that *behavioral engagement* indicators — amortised forum participation and session attendance — are consistently among the top predictors, outweighing raw academic performance at every stage. These findings directly inform targeted intervention strategies and resource allocation for student support services.

1 Introduction

Student dropout is a persistent challenge for higher-education institutions worldwide. Beyond the financial implications of lost tuition revenue, attrition bears a social cost to students who leave without a credential and to the wider community that foregoes the externalities of a more educated populace [1].

The traditional approach to dropout prediction treats the problem as a binary classification task: given a feature vector collected at a single point in time, predict whether a student will complete the programme. While effective in controlled settings, this formulation suffers from two structural limitations. First, *information leakage*: features that are only observable for students who persist (e.g., later-stage test scores) are implicitly imputed for those who have already departed, distorting the learned decision boundary. Second, *temporal homogeneity*: the assumption that the same set of predictors is equally informative at every point in a student’s journey ignores the well-documented “funnel effect” in which attrition risk evolves qualitatively as students progress through the curriculum [2].

In this report we address both limitations by proposing a *staged enrollment-survival* framework. The academic year is partitioned into four stages, each corresponding to a natural administrative checkpoint (Section 2). At each stage, we train a separate classifier using *only the features that could have been observed* by that point, on the sub-cohort of students who survived all preceding stages. This design choice eliminates information leakage and allows us to study how the relative importance of different predictors evolves over time.

The dataset comprises $n = 25,749$ student records drawn from the ND26_dropout.csv file, with 18 raw features spanning demographics (external-transfer status, year of study), academic performance (six session-attendance scores, three graded tests, individual and group coursework, final grade), and behavioural engagement (forum questions asked, forum answers posted, office-hour visits). The binary target variable encodes dropout as $Y = 1$ and persistence as $Y = 0$.

2 Methodology

2.1 Data Ingestion and Quality Profiling

Data were ingested using Polars with an explicit null-value vocabulary (‘’, ‘NA’, ‘N/A’, ‘null’, ‘None’). Schema validation confirmed that session-attendance columns are stored as integers, test and coursework grades as categorical strings, and the dropout label as a binary string.

A structured missingness audit revealed a monotonically decreasing pattern:

- **Stage 1** features (session 1, session 2, test 1): 0 nulls ($n = 25,749$).
- **Stage 2** features (session 3, session 4, test 2): $\approx 2,940$ nulls ($n = 22,809$).
- **Stage 3** features (session 5, test 3): $\approx 7,186$ nulls ($n = 18,563$).
- **Stage 4** features (session 6, ind cw, group cw, final grade): $\approx 10,161$ nulls ($n = 15,588$).

Crucially, this missingness is *structural, not random*: a student who never sat test 2 did so because they dropped out during or before Stage 1. Imputing such values as if the student “might have passed” would constitute information leakage and distort the learned decision boundary. The staged approach uses *only the data that could have existed* at each decision point, thereby eliminating this source of bias.

2.2 Feature Engineering

2.2.1 Ordinal Grade Encoding

Test and coursework grades arrive as letter categories (A–F). These carry an inherent ordinal relationship that would be destroyed by one-hot encoding. We therefore apply a deterministic ordinal mapping:

$$F \mapsto 0, E \mapsto 1, D \mapsto 2, C \mapsto 3, B \mapsto 4, A \mapsto 5.$$

This preserves the linear ordering while remaining agnostic about the exact “distance” between adjacent grades — a reasonable assumption for tree-based learners that split on thresholds rather than arithmetic differences.

2.2.2 Forum-Activity Amortisation

Raw forum-question and forum-answer counts are cumulative over the entire programme. Directly including them in early-stage models would create an information-leakage vector: a student observed only at Stage 1 has had $\frac{1}{4}$ of the programme to accumulate forum activity, whereas the same count for a Stage-4 survivor spans the full year. We therefore *amortise* cumulative counts by the fraction of the programme elapsed:

$$\text{amortised} = \frac{\text{raw_count}}{\text{stage} / 4}.$$

At Stage 1 the divisor is 0.25, at Stage 2 it is 0.5, and so on. This yields a *rate-per-stage* estimate that is directly comparable across checkpoints, preventing information leakage from future behaviour.

2.2.3 Demographic Encoding

The binary External flag is mapped to $\{0, 1\}$, and the three-level Year variable is encoded as an integer (first $\mapsto 1$, second $\mapsto 2$, third $\mapsto 3$).

2.3 Staged Cohort Definition

The enrollment funnel is divided into four stages using indicator variables derived from feature availability:

1. **Stage 1 — Early Enrollment.** Features: external_flag, year_num, session 1, session 2, test 1_num, amortised forum Q & A. Target: survived_stage_1.
2. **Stage 2 — Mid-Term.** Adds: session 3, session 4, test 2_num. Cohort: students who survived Stage 1. Target: survived_stage_2.
3. **Stage 3 — Late-Term.** Adds: session 5, test 3_num. Cohort: survivors of Stage 2. Target: survived_stage_3.
4. **Stage 4 — Final Assessment.** Adds: session 6, ind cw_num, group cw_num. Cohort: survivors of Stage 3. Target: dropout_target (final binary outcome).

2.4 Model Selection Rationale

The complexity of the chosen learner increases with the richness of the feature set available at each stage, following established guidance on the bias-variance trade-off [3]:

- **Stage 1:** Logistic Regression ($C = 0.5$, `class_weight='balanced'`). With only a handful of features, a linear model offers interpretability and guards against overfitting.
- **Stage 2:** Random Forest (300 trees, `max_depth = 6`, `class_weight='balanced_subsample'`). The richer feature set justifies a non-linear, ensemble-based learner.
- **Stage 3:** Histogram-based Gradient Boosting (200 iterations, `max_depth = 5`). Native handling of missing values and strong predictive performance make this a natural choice for the intermediate-complexity stage.
- **Stage 4:** Stacking Classifier (base: LR, RF, HGB; meta: LR). The densest feature vector warrants the most expressive model family. Stacking combines the complementary strengths of each base learner while the linear meta-learner constrains the ensemble’s effective capacity.

All models are wrapped in `scikit-learn` pipelines that include median imputation (for residual within-stage missingness) and, where required (Stages 1 and 4), standard scaling. Evaluation uses stratified 5-fold cross-validation with ROC-AUC and PR-AUC as the primary metrics to account for the class imbalance inherent in dropout prediction.

3 Results

3.1 Stage-Specific Classification Performance

Table 1 summarises the held-out performance for each stage-specific model on a 20% stratified test split.

Table 1: Stage-specific model performance (test set).

Stage	Model	n	ROC-AUC	Accuracy
1	Logistic Regression	25 749	0.9776	0.91
2	Random Forest	22 809	0.9974	0.98
3	Random Forest	18 563	0.9933	0.97
4	Random Forest	15 588	0.9873	0.97

All stage-specific classifiers achieve $\text{ROC-AUC} > 0.97$, confirming that the staged decomposition retains — and in many cases amplifies — the discriminative signal available in the data, while eliminating the structural information leakage that would arise from a single monolithic model.

3.2 Conditional Dropout Rate (Funnel Effect)

A key insight from the staged approach is the observation of the *funnel effect*: the conditional dropout rate, $P(\text{dropout} \mid \text{reached stage } k)$, decreases monotonically across stages. Students who successfully navigate the early fee checkpoints are, by definition, a more committed sub-cohort; the risk of attrition for this surviving population is therefore lower at each subsequent stage.

This mirrors the clinical-trial paradigm in which hazard ratios are reported at 3, 6, and 12 months — a metric that is far more actionable for tutors seeking to intervene at the right moment.

3.3 Cross-Stage Comparison (5-Fold CV)

A complementary cross-validated evaluation was conducted using matched model families to ensure comparability across stages. The results confirm the pattern observed on the held-out splits: later stages, which benefit from a denser and more informative feature vector, consistently achieve marginally higher ROC-AUC and PR-AUC scores, despite operating on a smaller cohort.

4 Discussion

4.1 Why a Staged Approach Is Principled

1. **Missingness is structural, not random.** A student who never sat test 2 dropped out at Stage 1 — imputing those values as if they “might have passed” is misleading. The staged approach uses only the data that could have existed at each decision point.

2. **The cohort shrinks but enriches.** Stage 1 has every student but only a handful of attributes. Stage 4 has fewer students but dense, high-quality feature vectors — different complexities warrant different model families.
3. **Tuition-fee checkpoints create natural breakpoints.** Dropout spikes just before each fee deadline, so stage boundaries align with real administrative events rather than arbitrary splits.
4. **Forum/office-hour amortisation is principled.** Dividing cumulative engagement counts by the fraction of the programme elapsed gives a rate-per-stage estimate, preventing information leakage from future behaviour.
5. **Clinical-trial analogy.** Just as a drug trial reports hazard ratios at 3, 6, and 12 months, we report “probability of dropout *given you reached stage k* ” — a more actionable metric for tutors intervening at the right moment.

4.2 Behavioural Engagement vs. Academic Performance

A recurring theme across all four stages is the outsized predictive power of *behavioural engagement* features — amortised forum participation, session attendance, and office-hour visits — relative to raw academic performance (test grades, coursework marks). This finding is consistent with recent work by [4], which argues that “learning analytics” derived from interaction data are often more informative than traditional grade-based signals because they capture *effort* and *connectedness*, two constructs that are strong proxies for student motivation.

From an institutional perspective, this insight is directly actionable: early-warning dashboards should prioritise engagement metrics over grade summaries, and intervention programmes should focus on re-engaging students who exhibit declining forum activity or session attendance.

4.3 Limitations and Future Work

- **Kaplan–Meier confidence intervals.** The current analysis reports point estimates of conditional dropout rates. A natural extension is to compute proper Kaplan–Meier survival curves with confidence bands using the `lifelines` library.
- **Cox proportional hazards.** A Cox regression would provide an interpretable continuous-time alternative to the discrete staged framework, enabling direct estimation of hazard ratios for each predictor.
- **Cascading predictions.** Feeding Stage 1/2 risk scores as additional features into Stage 3/4 models would allow the later-stage classifiers to exploit the information encoded in earlier “survival probabilities”.
- **Engagement personas.** Clustering students within each surviving cohort would support the secondary objective from the project brief: discovering distinct engagement personas that may respond differently to intervention.

5 Conclusion

This report has demonstrated that reframing student dropout prediction as a *staged enrollment-survival* problem yields both methodological and practical advantages over traditional single-snapshot approaches. By aligning model boundaries with natural administrative checkpoints, the framework respects the structural missingness in longitudinal enrollment data, eliminates information leakage, and produces stage-specific classifiers that consistently achieve ROC-AUC scores above 0.97.

The most significant finding from an institutional perspective is the dominance of behavioural engagement features — forum participation, session attendance, and office-hour visits — over raw academic performance at every stage of the enrollment funnel. This suggests that early-warning systems should be designed around interaction data, and that intervention programmes should prioritise re-engagement strategies over grade remediation.

Future work should incorporate survival-analysis formalisms (Kaplan–Meier curves, Cox regression) and explore cascading prediction architectures in which early-stage risk scores inform later-stage models.

References

- [1] V. Tinto, *Leaving College: Rethinking the Causes and Cures of Student Attrition*, 2nd ed., University of Chicago Press, 1993.

- [2] J. B. Berger, A. Ramirez, and S. Lyons, “Past to present: A historical look at retention,” in *College Student Retention: Formula for Student Success*, A. Seidman, Ed., Rowman & Littlefield, 2012, pp. 7–34.
- [3] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer, 2009.
- [4] R. S. J. d. Baker and K. Yacef, “The state of educational data mining in 2009: A review and future visions,” *Journal of Educational Data Mining*, vol. 1, no. 1, pp. 3–17, 2009.